

Muzzammil Ahmed

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🌐 [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

📍 Bhopal, Madhya Pradesh, India

PROFESSIONAL SUMMARY

M.Tech Computer Science Engineering student with hands-on experience across Data Analytics and Machine Learning. Skilled in data cleaning, exploratory data analysis, statistical analysis, dashboarding, and deriving actionable business insights, as well as building and deploying ML/DL models including CNN-based computer vision and RAG-based NLP solutions. Comfortable working across the full pipeline from raw data to predictive models to interactive Streamlit/BI applications. Currently conducting research on hybrid machine learning models for heart disease prediction

EDUCATION

All Saints' College of Technology <i>M.Tech in Computer Science Engineering</i> CGPA: 7.63	Aug 2024 – Expected Dec 2026 Bhopal, India
All Saints' College of Technology <i>B.Tech in Computer Science Engineering</i> CGPA: 8.32	Jun 2020 – Jul 2024 Bhopal, India
Royal Higher Secondary School <i>Higher Secondary (PCM)</i> 63%	2018 – 2020 Bhopal, India
Kendriya Vidyalaya No.3 <i>High School</i> 62%	2008 – 2018 Bhopal, India

Skills

Data Analytics: Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization, Statistical Analysis, Feature Engineering

Business Intelligence: Power BI, Excel, Pivot Tables, Charts, Slicers, KPIs, Dashboard Development

Programming: Python, SQL, JavaScript

Databases: MySQL

Machine Learning: Regression, Classification, Decision Trees, KNN, Ensemble Learning, Feature Selection, Hyperparameter Tuning, Cross-Validation, SMOTE

Deep Learning: ANN, CNN

Artificial Intelligence: NLP, Transformers, RAG, LangChain

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, OpenCV, Streamlit, Matplotlib, Seaborn

Frontend Development: HTML, CSS, React.js

Tools: Git, GitHub, Jupyter Notebook, VS Code, Figma, Netlify

EXPERIENCE

Muslim Education and Career Promotion Society (MECAPS) <i>AI/ML Training Intern</i>	Jul 2025 – Jun 2026 Bhopal, India
• Worked on Machine Learning and Deep Learning concepts through practical training and project-based learning. • Developed ML models using Python, Pandas, NumPy, Scikit-learn, TensorFlow, and Keras. • Performed data preprocessing, exploratory data analysis, feature selection, model training, and model evaluation. • Worked with ANN, CNN, NLP, and hybrid machine learning approaches on real-world datasets.	

RESEARCH

Concatenated Hybrid Ensemble Model for Heart Disease Prediction *Under Review*

- Designed 5 hybrid architectures and 14 concatenated ensemble combinations using Logistic Regression and Decision Tree as base learners, inspired by concatenated hybrid ensemble voting approaches in literature.
- Best single-split model achieved 91.80% accuracy, 92.54% F1-score, 91.18% precision, and 93.94% recall on the Cleveland heart disease dataset.
- Implemented a leakage-free 10-fold cross-validation protocol with McNemar's test and repeated 5x2 cross-validation for statistical rigor; best cross-validated model (Extra-Trees-selected hybrid) achieved 85.41% accuracy, 87.22% F1-score, and 90.66% AUC.

PROJECTS

PDF Chatbot using RAG and LangChain [🔗 GitHub](#)

- Developed a Retrieval-Augmented Generation (RAG) chatbot for question answering over uploaded PDF documents.
- Implemented document loading, text chunking, embeddings, vector storage, and semantic retrieval for relevant context.
- Integrated LangChain with an LLM to generate context-aware responses based on retrieved document content.
- Implemented a Streamlit interface for interactive document-based question answering.

Delivery Time Prediction System [🔗 GitHub](#)

- Developed an end-to-end delivery time prediction system using the Porter logistics dataset.
- Performed exploratory data analysis, feature engineering, data preprocessing, and statistical analysis to identify key factors affecting delivery time.
- Built and optimized an ANN regression model using TensorFlow with hyperparameter tuning and model evaluation.
- Developed a Streamlit web application for real-time delivery time prediction using the trained model.

Face Mask Detection using CNN [🔗 GitHub](#)

- Developed a Convolutional Neural Network (CNN) model for classifying images as with-mask or without-mask.
- Performed image preprocessing and data preparation using OpenCV and NumPy for model training.
- Trained and evaluated the CNN model using TensorFlow and Keras for image classification.
- Built a Streamlit application for real-time face mask detection using the trained CNN model.

CERTIFICATIONS

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| • Python - Introduction to Data Science and Machine Learning A-Z | View Certificate |
| • Python for Data Analytics - Beginner to Advanced | View Certificate |
| • Learn MySQL - For Beginners | View Certificate |
| • Machine Learning Basics - Regression Analysis | View Certificate |

ADDITIONAL INFORMATION

Languages: English, Hindi

Areas of Interest: Artificial Intelligence, Machine Learning, Data Analytics, Natural Language Processing, Computer Vision

Strengths: Problem Solving, Analytical Thinking, Fast Learner, Research-Oriented Mindset, Adaptability